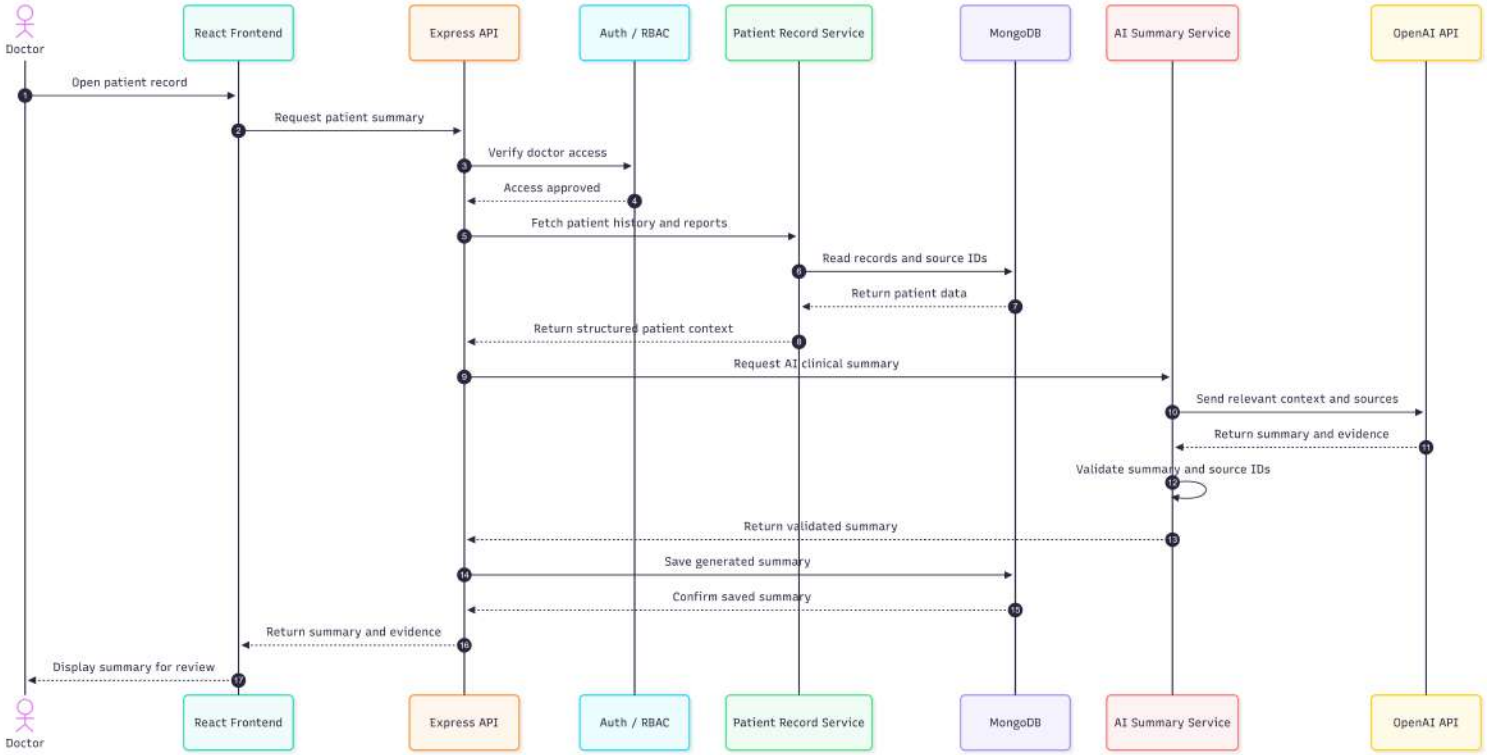

Start
Doctor logs in
Open patient record
Upload medical report
Validate file type and size
File valid?
No
Yes
Reject upload and show error
Store original document
Extract text from report
Text readable?
No
Yes
Run OCR on report
Normalize report values
Save observations with source IDs
Detect trends and record conflicts
Doctor requests AI summary
Build authorized patient context
Generate AI clinical summary
Validate summary and evidence
Display summary for doctor review
End


Doctor
React Frontend
Express API
Auth / RBAC
Patient Record Service
MongoDB
AI Summary Service
OpenAI API
1 Open patient record
2 Request patient summary
3 Verify doctor access
4 Access approved
5 Fetch patient history and reports
6 Read records and source IDs
7 Return patient data
8 Return structured patient context
9 Request AI clinical summary
10 Send relevant context and sources
11 Return summary and evidence
12 Validate summary and source IDs
13 Return validated summary
14 Save generated summary
15 Confirm saved summary
16 Return summary and evidence
17 Display summary for review
Doctor
React Frontend
Express API
Auth / RBAC
Patient Record Service
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AI Summary Service
OpenAI API